SYSTEMS & DEVICES 3 (SYS3 - COM00018I) FORMATIVE EXERCISE 2

Covering Control Hazards, Branch Prediction, Static and Dynamic Scheduling, Superscalar Processors.

The questions for the assignment exercises are partly adapted from the book - Computer Architecture: A Quantitative Approach, 5th Edition, John L. Hennessy & David A Patterson

Please try to complete it by 27/03/2026 (Fri, S2 Week 7). We will use the Lecture Review session on 16/04/2026 (Thu, S2 Week 8) for discussions if there are any questions on it.

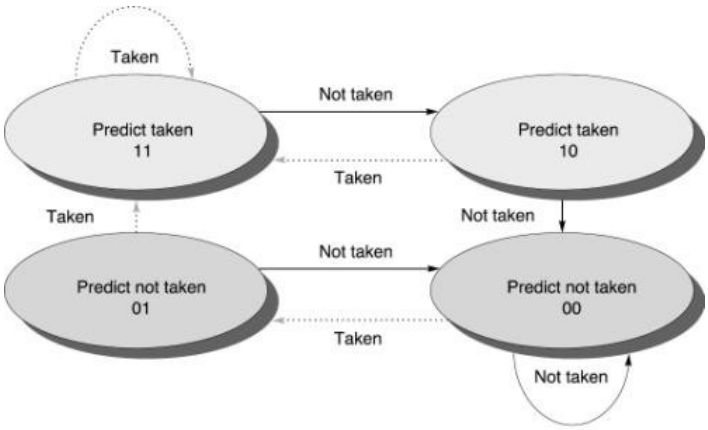
Q1. Explain what is Delayed Branching?

Q2. True or False: In a pipelined processor, control hazard penalties can be eliminated by dynamic branch prediction.

Q3. We have a program core consisting of five conditional branches. The program core will be executed thousands of times. Below are the outcomes of each branch for one execution of the program core (T for taken, N for not taken).
Branch 1: T-T-T-T
Branch 2: N-N-N-N
Branch 3: N-T-N-T-N-T
Branch 4: T-N-T-N-T-N
Branch 5: T-T-N-T-T-N-T-T

Assume the behaviour of each branch remains the same for each program core execution. For the dynamic branch prediction schemes below, assume that each branch maps to a unique entry in the branch history table. Further assume that each BHT entry is initialized to the same stage before each execution

(a) Always taken
(b) Always not taken
(c) 1-bit predictor, initialized to predict taken
(d) 2-bit predictor (Finite State Machine as shown below), initialized to predict taken (10)



What is the branch prediction accuracy for branch 1 to 5 with prediction scheme (a) through (d)? Use the following table to fill out your final answer.

Branches	Prediction Schemes			
	(a)	(b)	(c)	(d)
1				
2				
3				
4				
5				

Q4. What is the shortcoming of Tomasulo's algorithm that leads to the introduction of speculation? How is the reorder buffer used to introduce speculation?

Q5. (True or False) In super-scalar mode, all the similar instructions are grouped and executed together.

Q6. In super-scalar processors, out of order mode of execution is used, the results are stored in _____?

a) Buffers

b) Special memory locations

c) Temporary registers

d) TLB

Q7. A special unit used to govern the out of order execution of the instructions is called as _____

a) Commitment unit

b) Temporal unit

c) Monitor

d) Supervisory unit

Q8. The commitment unit uses a queue called _____

a) Reorder buffer

b) Commitment buffer

c) Storage buffer

d) None of the mentioned

Q9. Consider the following instruction sequence:

1. LD R1, 0(R2)

2. ADD R3, R1, R4

3. SUB R5, R3, R6

4. MUL R7, R8, R9

5. ADD R10, R1, R11

Assume: Load latency = 2 cycles; ADD/SUB latency = 1 cycle; MUL latency = 3 cycles; In-order issue pipeline.

Answer the following questions:

a) Identify data dependencies

b) Perform static scheduling to minimize stalls

c) Draw the execution timeline

Q10. Given the loop:

Loop:

LD F0, 0(R1)

ADD F4, F0, F2

SD F4, 0(R1)

ADDI R1, R1, #8

BNE R1, R3, Loop

Assume: Load latency = 2 cycles; ADD latency = 2 cycles; Store latency = 1 cycle.

Answer the following questions:

a) Unroll the loop 2 times.

b) Apply static scheduling to reduce stalls.

Q11. Explain why the following pair cannot issue together:

1 ADD R1,R2,R3

2 SUB R4,R1,R5

Q12. Explain how superscalar processors exploit instruction-level parallelism (ILP) and discuss two major challenges.